

Efficient plasticizers for polyvinyl chloride

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Abstract

This thesis is devoted to the production and study of effective and environmentally friendly plasticizers using heterogeneous catalysts.

Introduction: Esters of aliphatic, alicyclic, and aromatic carboxylic acids are promising chemical products with wide practical use in various industrial fields as plasticizers. Plasticizers are among the most important classes of polymer additives that improve the plasticizing properties of materials. Since plasticizers are widely used in the production of consumer goods, it is important to consider their effects on human health. Currently, the use and market of plasticizers are under strict control of environmental and health protection directives. According to research, products made of PVC with the plasticizer dioctyl phthalate (DOP) are harmful to health, leading to bans on the production and sale of children's toys and some household items made from these materials [1].

Methods: We developed single-stage processes for obtaining esters of carboxylic acids using a heterogeneous catalyst (ZnO, TiO₂ (+FemOn)) and proposed a basic technological scheme [2]. Due to the advantages of carboxylic acid esters and to increase the range of effective, eco-friendly plasticizers, in this study we synthesized them from individual aromatic, aliphatic and alicyclic, natural, and synthetic petroleum-based carboxylic acids with alcohols in the presence of a heterogeneous catalyst. The synthesis was conducted in a reaction flask equipped with a mechanical stirrer, reflux condenser, water separator, and thermometer. The acid and an excess of alcohol (30%) were reacted in an organic solvent in the presence of heterogeneous catalysts until water release ceased. The physical and chemical parameters of the synthesized esters were determined to GOST 8728-88 standards.

Results: Besides suitable properties, a plasticizer must be well compatible with the polymer matrix. Therefore, the mixtures were thermostated for 3–6 hours at 65, 75, and 85°C until complete swelling of the PVC composition in the plasticizer. The mixtures were cooled to room temperature and held under load until greasy spots disappeared on filter paper, after which the optimal compatibility limit of the synthesized esters with PVC (30–40 parts by mass) was determined. Compositions with synthesized esters matured within 0.25–6 hours, while those with DOP matured in 8 hours, indicating better compatibility of the synthesized diesters with PVC compared to DOP.

Conclusion: The proposed synthesis scheme reduces the steps of neutralization, washing, and drying of the target products in accordance with global manufacturing standards, and significantly reduces corrosion in metal equipment and reactors. The scheme is distinguished by not requiring additional disposal of acidic and alkaline waste from the esterification process that uses acid catalysts. The use of heterogeneous catalysts simplifies the technology, thus ensuring environmental cleanliness.

Keywords: carboxylic acids, esterification, catalyst, plasticizer, polyvinyl chloride.

References

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